

derived cluster to the newly created base cluster.

NOTE

If a cluster is derived from an already derived cluster, or a new base cluster is created for an existing cluster, it is important to ensure there is no overlap in the defined ranges of the base and derived clusters. It also means that a derived or base cluster MAY contain elements which are outside of the proposed ranges. In that case, the clusters SHALL define custom ranges which fit into the already defined elements.

If an endpoint supports multiple server clusters that derive or map to the same base cluster specification, then each SHALL represent a single implementation and operate as a single entity or instance. This makes it possible to deploy a new device endpoint with both a base and a derived cluster identifier, which SHALL remain backward compatible to legacy devices that support only the original cluster identifier. Cluster identifiers that are mapped to a single base cluster specification, but are defined for distinctly different purposes, MAY exist together on a device endpoint. If there is no base cluster identifier defined, or no base cluster identifier exists on the same endpoint, then each cluster identifier SHALL represent a separate instance.

It is a good practice to explore the possibility of either deriving a cluster from an existing cluster or creating a base cluster to map or derive new and existing cluster identifiers. See [New Cluster](#) for other options.

7.10.7. Status Codes

A cluster specification defines status code responses to actions depending on the cluster instance state. A status code is either a [global Interaction Model status code](#), or a cluster specific status code that is unique to the cluster specification. A global status code is either scoped to the entire action, or to a cluster request path. A cluster specific status code scoped to a cluster instance is indicated by a cluster path. When an interaction defines a Status Response response, the responder SHALL return a global Interaction Model status code. When an interaction response needs to communicate a cluster specific status code, the responder SHALL return the path to the cluster instance, the global status code SUCCESS or FAILURE, and the cluster specific status code. Each cluster specific status code SHALL be associated with either SUCCESS or FAILURE, not both. A cluster specific status code SHALL be, by default, associated with FAILURE unless it is defined as associated with SUCCESS. The global SUCCESS status code means the action was executed for the request path; the global FAILURE status code means that it was not executed.

- Cluster-specific status code SHALL be defined using the [status](#) type.
- Cluster-specific status codes MAY have the same numeric values as global status codes. Interaction model messages SHALL make it clear whether a particular message field is a global status code or a cluster-specific status code.
- Cluster-specific status codes SHALL communicate more information than just a generic success or failure condition. Global status codes SHALL be used to communicate such conditions.
- A server cluster SHALL NOT return a cluster-specific code from another cluster.